

DAPO Decoupled clipping & dynamic sampling for long-CoT RL

Sample reasoning groups

Math prompt + answer
q : 6 × 7 = ? a : 42

Rollout snapshot
 $\pi_{\theta_{\text{old}}}$

6 completions / prompt
Fixed during collection

Dynamic sampling
1 0 1 Mixed → buffer

Token-level clipped objective

Clip-Higher
0.80 1 1.20 1.28
Raise the upper clip; keep the lower clip
Clip range shapes the surrogate objective.

Verify, shape & select

Completions

... 42 <eos>	+1	✓
... 17 <eos>	-1	✗
... 42 <eos>	+1	✓

Correctness + soft length reward

Retain groups with $0 < \text{accuracy} < 1$

All correct / all wrong → discard

Token-level reduction

Shared advantage; equal token weight

+ + +
- - - - -
m = 0 for truncated responses

Group signal & trainable policy

Normalize rewards within each retained group
 $\hat{A}_{i,t} = \hat{A}_i = \frac{R_i - \text{mean}(R)}{\text{std}(R)}$

One response advantage → all its valid tokens

Current policy
 π_{θ} Maximize the DAPO objective
No learned critic; no reference KL

Resample until the retained batch is full

$$\hat{J}(\theta) = \frac{\sum_{i,t} m_{i,t} \ell_{i,t}}{\sum_{i,t} m_{i,t}}, \quad \ell_{i,t} = \min(w_{i,t} \hat{A}_i, \text{clip}(w_{i,t}, 1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}) \hat{A}_i)$$

Soft overlong punishment

Correctness + length penalty

